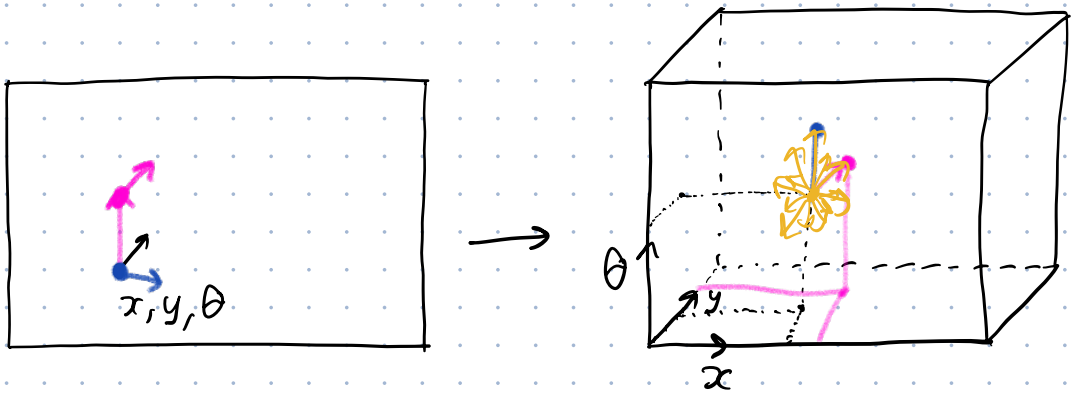
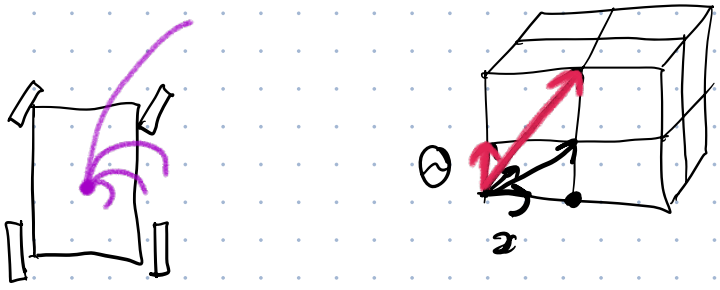
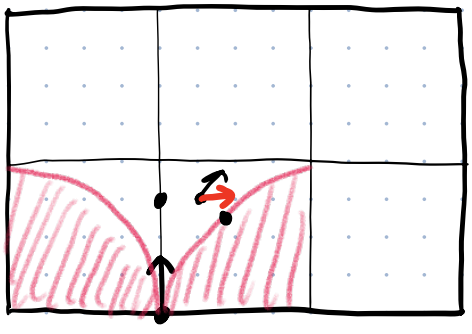
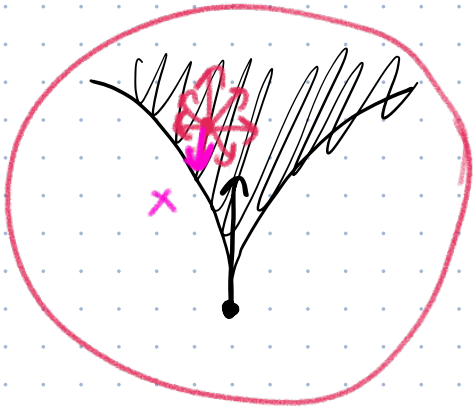
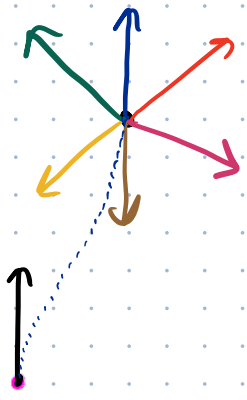
27. 03. 26

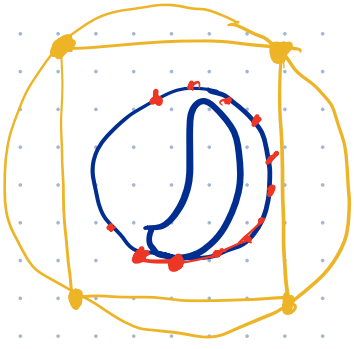


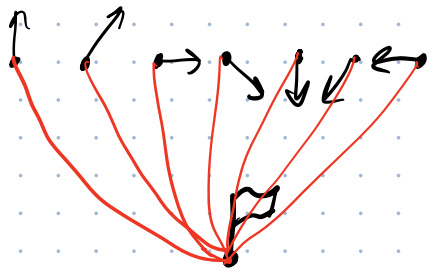


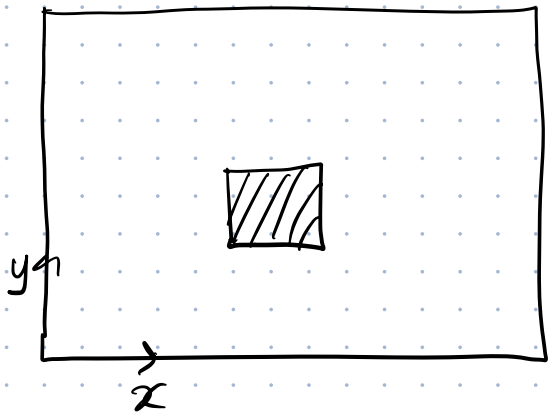


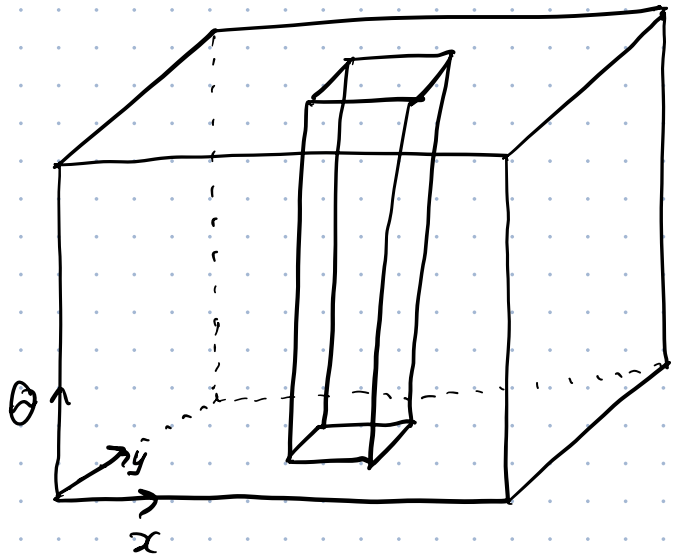












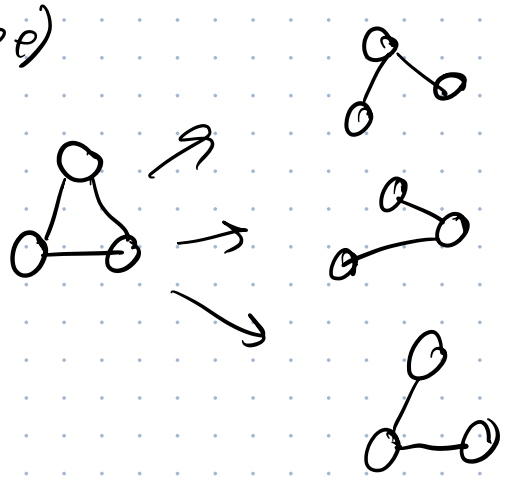
ОСТОВНЫЕ ДЕРЕВЬЯ

ДЕРЕВО – СВ. БЕЗ ЦИКЛОВ

∃ ПОДГРАФЫ НА ВСЕХ ВЕРШ.

ПОДГР. ГР. G НА ВС. ВЕРШ., ЯВЛ. ДЕР., НАЗ. ОСТОВНЫМ ДЕРЕВОМ (spanning tree)

ОНИ НЕ ЕДИНСТВ.:



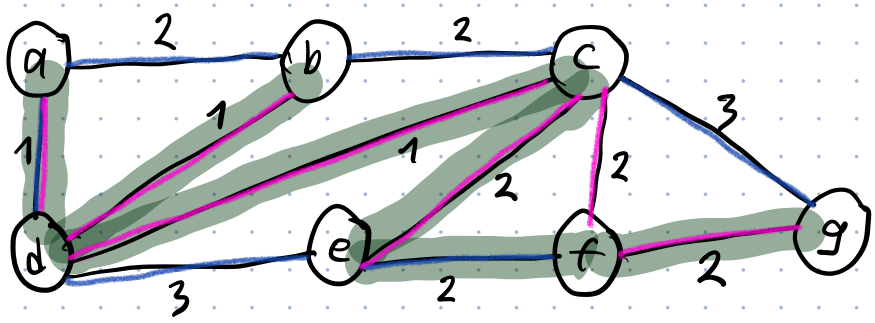
$ST(G) = G$

ДЛЯ ВЗВЕШ. ГР. ВЕС ОСТ. ДЕРЕВА – ЭТО СУММА ВЕСОВ РЁБЕР

MST - minimal spanning tree

АЛГ. КРАСКАЛА

1) СОРТ. РЁБРА ПО ВЕСУ
2) В ЦИКЛЕ ПОКА НЕ ДОБАВЛ. $|V|-1$ РЕБРО:
   ЕСЛИ ДОБАВЛ. ОЧЕРЕДН. РЕБРА НЕ ПРИВОДИТ К ПОЯВЛ. ЦИКЛОВ, ДОБАВЛЯЕМ



14

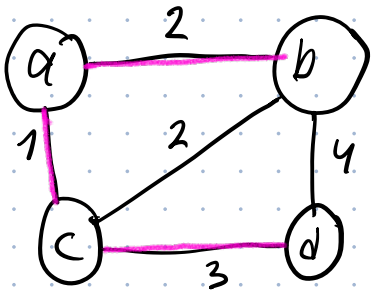
9

$\underbrace{db, da, dc}_{1}$, $\underbrace{\cancel{ab}, \cancel{bc}, ef, ec, \cancel{fc}, fg}$, $\boxed{de, cg}$

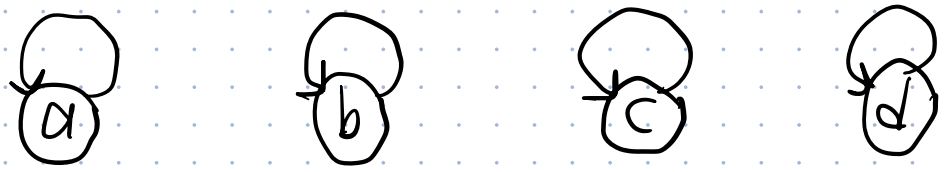
КАК ПРОВЕРЯТЬ НА ОБРАЗОВАНИЕ ЦИКЛОВ?
МОЖНО ДЕЛАТЬ dfs/bfs С ИСПОЛЬЗ. ТОЛЬКО ТЕХ РЁБЕР, КОТ. УЖЕ ДОБАВЛЕНЫ

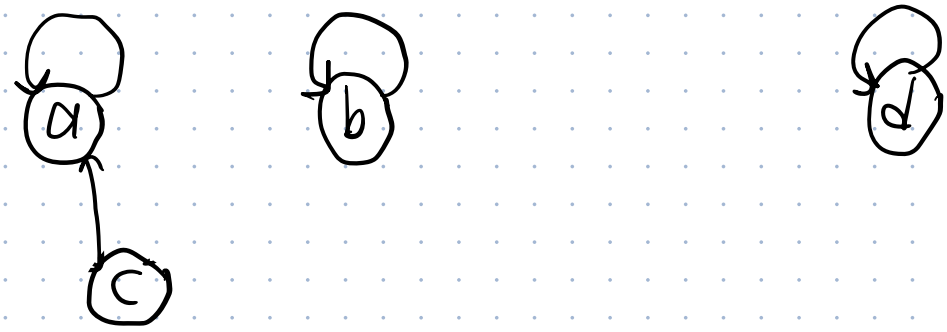
СЛОЖНОСТЬ: $O((|V|-1)(|V|+|E|))$

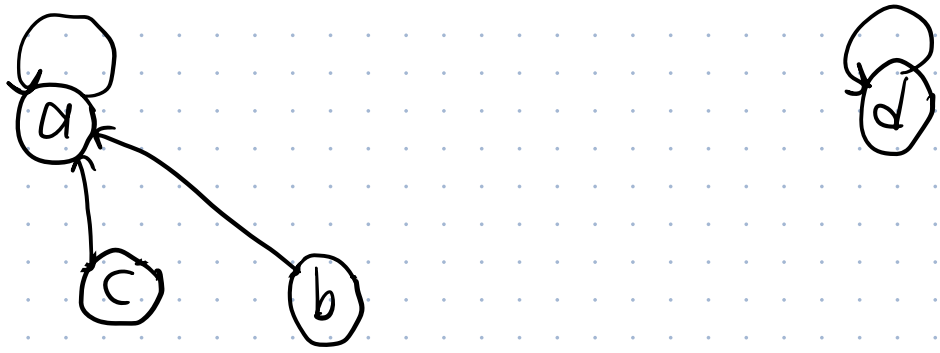
$O(|V||E|)$

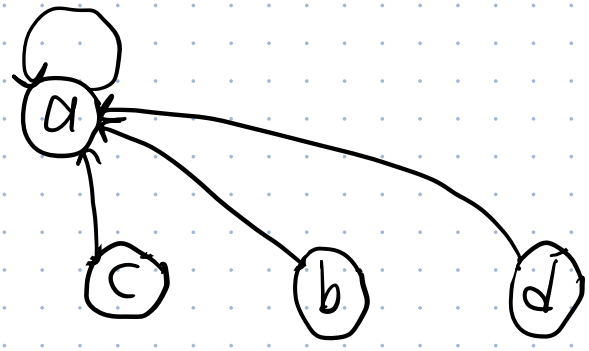


$\underbrace{ac}_{1}, \underbrace{ab, \cancel{cb}}_{2}, \underbrace{cd}_{3} \mid \underbrace{bd}_{4}$

1) 

2) 

3) 

4) 

СТР. ДАННЫХ disjoint set / union-find / СИСТ. НЕПЕРЕСЕК. МН.

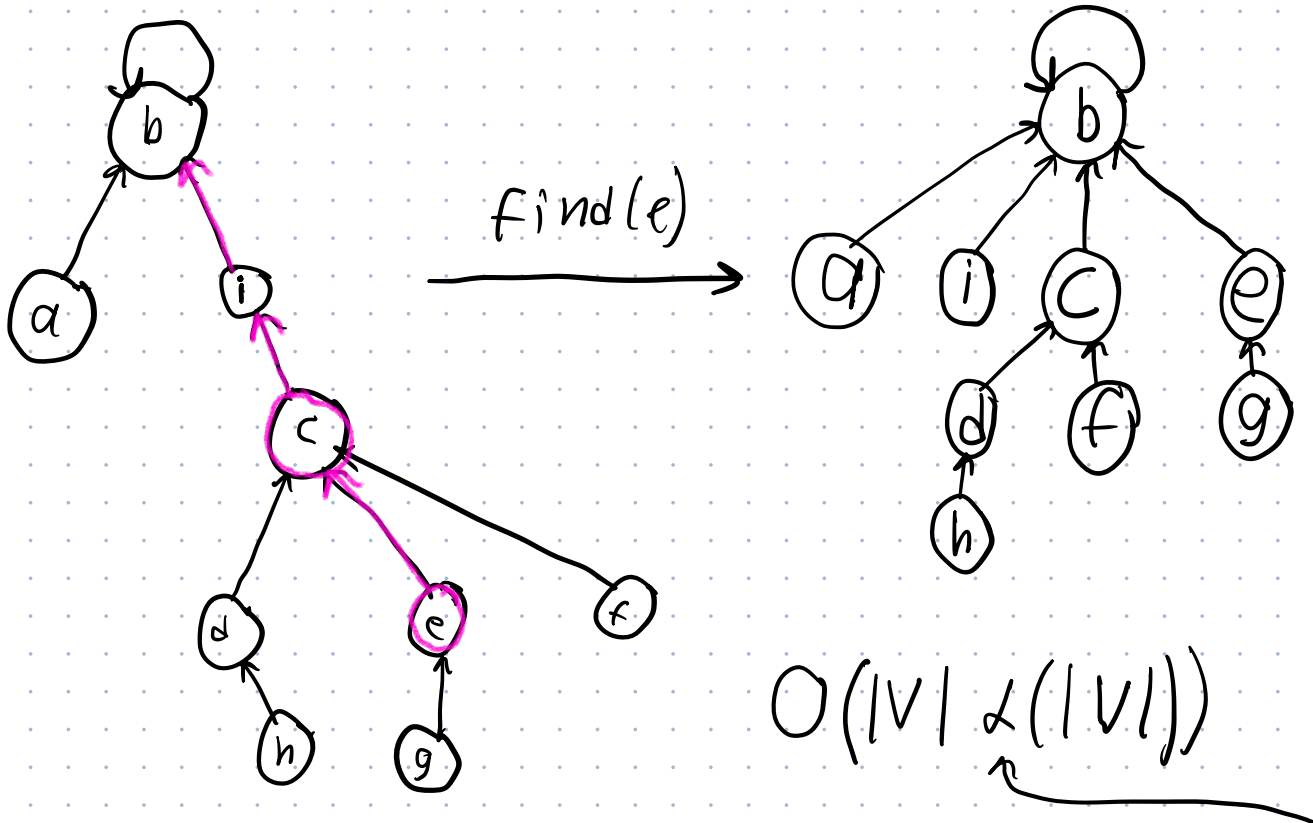
1) makeset (x) $O(1)$

2) find (x) $O(h)$

3) union (a, b)

1 ЭВРИСТИКА: ПОДВЕШ. БОЛЕЕ НИЗ. ПОД БОЛЕЕ ВЫС.

2 ЭВРИСТИКА (СОКРАЩЕНИЕ ПУТЕЙ):



$$O(|V|\,\alpha(|V|))$$

ОБРАТНАЯ Ф-Я АККЕРМАНА

ИТЕРИРОВАННЫЙ ЛОГАРИФМ – СКОЛЬКО РАЗ НУЖНО ПРИМ. $\log_2$ ЧТОБЫ ПОЛУЧ. ЧИСЛО < 2

$i(2) = 1$

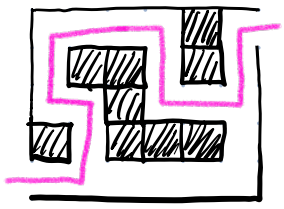
$i(2^2) = 2$

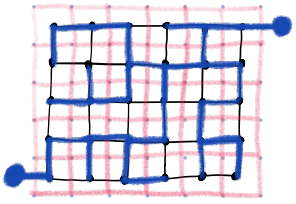
$i(2^{2^2}) = 3$

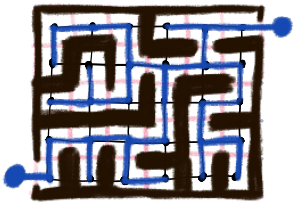
$i(2^{2^{2^2}}) = 4$

$$\alpha(n) = O(i(i(n)))$$

ЛАБИРИНТЫ







ДЗ:

КРАСКАЛ + ЛАБИРИНТ